



AI-DRIVEN

CYBER SECURITY

AN AI CHATBOT WAITS FOR YOUR QUESTION. AN AI AGENT TAKES A GOAL, THINKS ABOUT WHAT TO DO, USES TOOLS, REMEMBERS CONTEXT, AND COMPLETES THE TASK AUTONOMOUSLY.

WHAT IS AN AI AGENT?

- AI AGENT = GOAL + REASONING + TOOLS +
MEMORY
- WORKS TOWARD A SPECIFIC GOAL
- DECIDES THE NEXT ACTION
- CAN INTERACT WITH EXTERNAL SYSTEMS
- CAN COMPLETE MULTI-STEP TASKS

AUTONOMOUS WORKFLOWS

- TRADITIONAL AI: INPUT → RESPONSE
- AGENT: GOAL → PLAN → ACT → OBSERVE → REPEAT
- MAKES DECISIONS DURING EXECUTION
- DOESN'T NEED EVERY STEP PREDEFINED
- USEFUL FOR COMPLEX AUTOMATION

TOOL CALLING

- AGENTS CAN USE EXTERNAL TOOLS
- EXAMPLES: APIS, DATABASES, SEARCH, CALCULATORS
- AI DECIDES WHICH TOOL TO USE
- TOOL PERFORMS THE ACTION
- RESULT GOES BACK TO THE AI

MEMORY

- MEMORY STORES USEFUL INFORMATION
- SHORT-TERM: CURRENT CONVERSATION/TASK
- LONG-TERM: INFORMATION ACROSS SESSIONS
- HELPS AGENTS MAINTAIN CONTEXT
- MAKES INTERACTIONS MORE PERSONALIZED

PLANNING

- AGENTS BREAK A BIG GOAL INTO SMALLER TASKS
- DECIDE WHAT SHOULD HAPPEN FIRST
- EXECUTE TASKS STEP-BY-STEP
- ADJUST THE PLAN WHEN SOMETHING FAILS

MULTI-STEP REASONING

- COMPLEX TASKS NEED MULTIPLE DECISIONS
- AGENT PROCESSES ONE STEP AT A TIME
- EACH RESULT CAN INFLUENCE THE NEXT STEP
- TOOLS CAN BE USED BETWEEN STEPS

AGENT LOOP

- THE CORE IDEA BEHIND MANY AGENTS:
- GOAL
- THINK
- CHOOSE ACTION
- USE TOOL
- OBSERVE RESULT
- THINK AGAIN
- REPEAT
- DONE